



Basic II Training Project Manufacturing Plan

Name: Pranav Jain

STEP #	PROCESS DESCRIPTION	MACHINE	FIXTURE(S)	TOOL(S)	SPEED (RPM)	NOTES
1	Top Part: Mark sheet metal			Calipers		1" x 6.5" x 0.06"
2	Top Part: Cut sheet metal	Sheet Metal Shear				Align with bottom blade
3	Top Part: Mark Center Hole(s)			1) Calipers 2) Center Punch and Hammer		Pencil holes: 3/3.5/4" hole location, 0.5" edge Screw holes (x 2): 0.25" on either side, 0.25" edge
4	Top Part: Cut the Holes	Punch		Die Holder, Punch Holder, Spacer, 2 Wrenches, Stripper, 3/8" Punch, 3/8" Die		3/8" thru holes (Pencil holes) 1/4" thru holes (Screw holes) 1) Put Die in Die Holder 2) Put Punch in Punch Holder 3) Put Punch Holder in Machine 4) Put Die Holder, Spacer, and lock piece 5) Check if Die Holder and Punch Holder collide 6) Apply Cutting Fluid to Piece 7) Align Piece with Punch Holder 8) Punch hole 9) Attach Stripper and release
5	Top Part: Deburr			File, Hole Deburring Tool		
6	Top Part: Mark for Bending			Calipers		2.5" on either side
7	Top Part: Bend	Hand Brake				Align with blade (Make sure it is

						square)
8	Base Part: Mark Bar Stock			Calipers		2" x 1" x 0.5"
9	Base Part: Cut Bar Stock	Horizontal Band Saw	Vice			1) Lock feed rate control valve 2) Lock material in vice 3) Align material with blade 4) Spin wheel to tighten vice 5) Align guide close to material 6) Green button for Coolant 7) Unlock feed rate control valve to cut
10	Base Part: Deburr			File		
11	Base Part: Mark Center Holes			1) Calipers 2) Center Punch and Hammer		Pencil holes: 0.5/1/1.5" hole location, 0.5" edge Screw holes (x 2): 0.25" on either side, 0.25" edge
12	Base Part: Cut Pilot Holes	Drill Press	Vice	#2 Center Drill, Chuck Key,	Fast: 750	1) Put piece in vice 2) Put Center Drill in Chuck 3) Tighten chuck with check key 4) Line up machine with hole 5) Use Cutting Fluid 6) Cut roughly halfway up the bevel of center drill
13	Base Part: Cut Blind Holes	Drill Press	Vice	#21 Drill Bit (Screw Holes), 3/16" Drill Bit (Pencil Holes), Chuck Key	Slow: 750	1) Put piece in vice 2) Put Drill Bit in Chuck 3) Tighten chuck with check key 4) Set Depth of Screw Holes: 0.75" Pencil Holes: 0.25" 5) Line up machine with hole 6) Use Cutting Fluid 7) Peck Drilling until depth
14	Base Part: Tapping		Vice	Tap Handle, Tap		1) Material in Vice (Make sure it is square) 2) Put Tap in Tap Handle

						3) Use Cutting Fluid 4) Do couple turns and fix until square 5) Forward 1 back off 1/2
15	Assembly					
16	Turn in					Name Unique name Team Name Graduation Year Put in grey "Project Turn In" bins
17						